

41.1

(a) Answers to be drafted by students in proper memo form.

Introduction:

Marginal cost is:

Direct labour

Direct materials

Variable expenses

3.00
3.50
2.25
8.75

As selling price of £10 exceeds marginal cost of £8.75 we should accept (but see below).*

Proof

Without new order

With new order

Direct labour

Direct materials

Indirect manufacturing costs

Variable

Fixed

Administration expenses

Selling and distribution expenses

Finance expenses

600,000
700,000
450,000
50,000
120,000
60,000
20,000
2,000,000
2,400,000
400,000
472,500
50,000
120,000
60,000
20,000
2,087,500

Revenue

Sales (2,400,000 + 100,000)

Profit

2,500,000
412,500

*Depends on how other things affected besides simple accounting calculation, such as whether there is sufficient spare manufacturing capacity to produce the extra units.

(b) For extra order:

Marginal costs per unit (see (a))

Depreciation (£3,000 p.a. ÷ 15,000)

Running costs (£6,000 p.a. ÷ 15,000)

Marginal costs per unit

8.75
0.20
0.40
9.35

As £9.35 is greater than selling price of £9.25 do NOT accept.

Appendix 2

41.3

Year 1

Revenue £120 × 2,400

Less Variable costs

Direct labour £8 × 3,200

Direct materials £12 × 3,200

Variable expenses £10 × 3,200

Total variable cost

Less in (A) Valuation closing inventory

800 × £96,000

3,200

Fixed indirect manufacturing costs

(A) Lima Ltd (Marginal) 288,000
(B) Delfina Ltd (Absorption) 288,000
25,600
38,400
32,000
96,000
24,000
72,000
160,000
256,000
64,000
232,000
56,000
432,000
32,000
48,000
40,000
120,000
24,000
144,000
36,000
108,000
160,000
344,000
84,000
26,000
164,000
260,000
172,000

Less in (B) Valuation closing inventory

800 × £256,000

Total costs

Gross profit

Year 2

Revenue £120 × 3,600

Less Variable costs

Direct labour £8 × 4,000

Direct materials £12 × 4,000

Variable expenses £10 × 4,000

Total variable cost

Add in (A) Opening inventory b/d

Add in (B) Opening inventory b/d

Less in (A) Valuation closing inventory

1,200 × 120,000

4,000

Fixed indirect manufacturing costs

Less in (B) Valuation of closing inventory

1,200 × (£120,000 + £160,000)

4,000

Total costs

Gross profit

41.3 (cont'd)

<i>Year 3</i>		
Revenue £120 × 3,000	(A) 360,000	(B) 360,000
Less Variable costs		
Direct labour £8 × 3,400	27,200	27,200
Direct materials £12 × 3,400	40,800	40,800
Var. expenses £10 × 3,400	34,000	34,000
Total variable cost	102,000	102,000
Add in (A) Op. inventory b/d	36,000	
Add in (B) Op. inventory b/d		84,000
	138,000	186,000
Less in (A) Valuation of closing inventory	48,000	
1,600 × 102,000	90,000	
3,400	160,000	
Fixed indirect manufacturing costs		160,000
		346,000
Less in (B) Valuation of closing inventory		123,294
1,600 × (102,000 + 160,000)		250,000
3,400		110,000
Total costs		222,706
Gross profit		137,294

41.4

(a) See text

(b) Lima Ltd has an inventory value of £48,000 in its statement of financial position; Delfina Ltd has an inventory valuation of £123,294. The difference of £75,294 is the difference in the overall gross profits of the two companies.

41.5

(a) Subject to points raised in (c) the extra production should be taken on, as this results in greater profits amounting to £1,562,000. Proof is as follows:

Extra revenue 60,000 × £150	9,000,000
Less Extra costs	
Direct materials (W1)	3,141,600
Direct labour 60,000 × 18.70 × 120% =	1,346,400
Variable overhead 60,000 × 7.50	450,000
Fixed costs	2,500,000
Extra profit	7,438,000
(W1) 60,000 × 74.80	1,562,000
Less Saving 10% on extra materials	4,488,000
Saving 10% on materials used on day shift	
10% × 120,000 × 74.80	897,600
	1,346,400
	3,141,600

(b) Breakeven point to justify night shift

Sale price per unit	150.00
Less Costs per unit	
Material	74.80
Direct labour 18.70 + 20%	22.44
Variable overhead	7.50
Contribution per unit	104.74
	45,236

Total fixed costs $\frac{2,500,000}{45.26}$ = Breakeven at 55,236 units

Note: No 10% reduction on materials because demand less than extra 60,000 units.

- (c) (i) Would firm be able to maintain selling price of £187 on first 120,000 units per year?
(ii) Would it have been more profitable to subcontract extra units needed?
(iii) Could we diversify into a more profitable alternative product?
(iv) Could extra day facilities have been more profitable?

41.8

(a)

Arncliffe Ltd			
Revenue Statement for the year ending . . .			
	Crowns	Kings	Total
Revenue	60,000	25,000	85,000
Direct costs:			
Raw mats	8,000	2,000	10,000
Labour	20,000	10,000	30,000
M/c running costs	12,000	3,000	15,000
Contribution	20,000	10,000	30,000
Rate of contribution to revenue	33.3%	40%	35.3%

41.8 (cont'd)

(b) Best product mix for next year:

Crowns manufactured per hour = $20,000/8,000 = 2.5$ per hourKings manufactured per hour = $10,000/2,000 = 5$ per hour

Kings gives best contribution rate, so produce Kings up to maximum requirements.

	Crowns		Kings	
	Units	Hours	Units	Hours
Minimum required	$6,000 \div 2.5$	2,400	$6,000 \div 5$	1,200
Produce up to maximum of 36,000 Kings		<u>2,400</u>	$30,000 \div 5$	<u>6,000</u>
				<u>7,200</u>
Still $(10,000 - 7,200 - 2,400) = 400$ hours left, left, so now produce Crowns	$1,000 \div 2.5$	<u>400</u>		<u>7,200</u>

Best mix is therefore: Crowns $(6,000 + 1,000) = 7,000$ Kings $(6,000 + 30,000) = 36,000$ Revenue therefore: $7,000 \times £3 = 21,000$; $36,000 \times £2.5 = 90,000$

Contributions:	Crowns (33.3%)	7,000
	Kings (40%)	<u>36,000</u>
Floor space costs		15,000
Insurance		<u>600</u>
Profit		<u>27,400</u>

(c) Product mix with extra machine

As maximum requirements for Kings have already been met in (b), all new output will be of Crowns.

Revenue (as before) in £	Crowns	Kings
Extra $(10,000 \text{ hours} \times 2.5) = 25,000 \times £3 =$	21,000	90,000
	<u>75,000</u>	<u>90,000</u>
Contributions: Crowns (33.3%)	32,000	
	Kings (40%)	<u>36,000</u>
Hire of extra machine		20,000
Floor space costs		15,000
Insurance		<u>600</u>
		<u>35,600</u>
		<u>32,400</u>

(d) Briefly:

(i) Market demand maintained.

(ii) Flexibility of return of extra machine if demand falls.

(iii) To see if wholesaler will guarantee minimum orders.

(iv) Are there outlets possible other than wholesaler?

(v) Selling price to wholesaler.

Appendix 2

41.10

(a) Paul Wagtail

Manufacturing Account and Statement of Profit or Loss for the year ending 30 April 2016

	Marginal method	Absorption method
Purchases of raw mats (125,000 – 2,100)	122,900	122,900
Carriage of raw materials	<u>1,500</u>	<u>1,500</u>
	124,400	124,400
Less Inventory raw materials	<u>8,900</u>	<u>8,900</u>
Cost of raw materials consumed	115,500	115,500
Production wages	<u>105,270</u>	<u>105,270</u>
Prime cost	220,770	220,770
Factory overhead expenses:		
Factory power	12,430	12,430
Factory supervisors' wages	29,600	29,600
Factory repairs	19,360	19,360
Factory insurance (40%)	1,920	1,920
Factory heating & light (40%)	1,440	1,440
Depreciation of plant	<u>17,600</u>	<u>17,600</u>
	82,350	82,350
	303,120	303,120
Less Work in progress (W1)	<u>11,038</u>	<u>11,038</u>
Production cost of goods completed c/d	292,082	287,964
Revenue	292,082	287,964
Less Inventory finished goods (W2)	<u>18,447</u>	<u>18,187</u>
Gross profit	273,635	269,777
Less Expenses:	<u>190,725</u>	<u>194,583</u>
Administration expenses	46,700	46,700
Distribution expenses	25,400	25,400
Selling expenses	23,800	23,800
Insurance (60%)	2,880	2,880
Heating and lighting (60%)	2,160	2,160
Depn: delivery vehicles	<u>35,200</u>	<u>35,200</u>
Net profit	136,140	136,140
	<u>54,585</u>	<u>58,443</u>

Workings:

(W1) $625 \times 80\% = 500$ equivalent making total $9,500 + 500 = 10,000$

Valuations: Marginal $\frac{500}{10,000} \times 220,770 = 11,038$

Absorption $\frac{500}{10,000} \times 303,120 = 15,156$

(W2) Marginal $\frac{600}{9,500} \times 292,082 = 18,447$

Absorption $\frac{600}{9,500} \times 287,964 = 18,187$

(b) See text.

41.14

(a)	K	L	M	N
Direct labour and materials	28	56	120	64
Variable overheads	8	16	26	24
Fixed overhead	4	8	14	12
Total cost per unit	40	80	160	100
Add Profit 20%	8	16	32	20
Selling price	48	96	192	120

(b) Discontinue M. The others are above marginal cost whereas M is below it.

(c)	(i) Followed our advice	(ii) Produced all items
Revenue	6,600	6,600
K 100 × £66	7,800	7,800
L 100 × £78	–	14,000
M 100 × £140	9,800	9,800
N 100 × £98	24,200	38,200

Less Costs

Direct labour and materials	14,800	26,800
(i) (28 + 56 + 64) × 100		
(ii) (28 + 56 + 120 + 64) × 100		
Variable overhead	4,800	7,400
(i) (8 + 16 + 24) × 100		
(ii) (8 + 16 + 26 + 24) × 100		
Fixed overhead	3,800	3,800
	23,400	38,000
Net profit	800	200

Note: The net profit or loss could have been worked out using contributions per items e.g. (i) Contributions per unit (i.e. Selling price less Marginal cost).

K 66 – (28 + 8) = 30	
L 78 – (56 + 16) = 6	
N 98 – (64 + 24) = 10	
Less Fixed costs	46 × 100 of each = 4,600
Net profit	3,800
	800

In (ii) the contribution from M would be negative.

(d) Discontinue K and N. All other items are above marginal cost.

(e)	(i) Followed our advice	(ii) Produced all items
Revenue K 100 × £34	–	3,400
L 100 × £96	9,600	9,600
M 100 × £280	28,000	28,000
N 100 × £78	–	7,800
	37,600	48,800

Less Costs

Direct labour and materials	17,600	26,800
(i) (56 + 120) × 100		
(ii) (28 + 56 + 120 + 64) × 100		
Variable costs	4,200	7,400
(i) (16 + 26) × 100		
(ii) (8 + 16 + 26 + 24) × 100		
Fixed overhead	3,800	3,800
	25,600	38,000
Net profit	12,000	10,800

42.1

	Production departments				Service departments		
	A	B	C	D	K	L	M
Indirect lab.	8,000	12,000	16,000	4,000	3,000	6,000	8,200
Other exp.	<u>5,400</u>	<u>6,200</u>	<u>7,200</u>	<u>3,000</u>	<u>9,000</u>	<u>4,000</u>	<u>4,000</u>
	13,400	18,200	23,200	7,000	12,000	10,000	12,200
Apportionment of costs:							
Dept K	2,400	3,000	3,600	1,800	(12,000)	-	1,200
Dept L	3,500	-	4,500	2,000	-	(10,000)	-
							<u>13,400</u>
Dept M	-	5,360	3,350	4,690	-	-	(13,400)
	<u>19,300</u>	<u>26,560</u>	<u>34,650</u>	<u>15,490</u>	-	-	-

(a) Overhead rates per direct labour hour

Department A	$\frac{19,300}{4,000} = £4.825$
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Department C	$\frac{34,650}{8,900} = £3.89$
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(b) Overhead rates per machine hour

Department B	$\frac{26,560}{5,200} = £5.11$
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Department D	$\frac{15,490}{4,800} = £3.23$
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